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Urban hubs of connectivity: contrasting patterns of gene flow within and among cities in the western black widow spider

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As urbanization drastically alters the natural landscape and generates novel habitats within cities, the potential for changes to gene flow for urban-dwelling species increases. The western black widow spider (*Latrodectus hesperus*) is a medically relevant urban adapter pest species, for which we have previously identified population genetic signatures consistent with urbanization facilitating gene flow, likely due to human-mediated transport. Here, in an analysis of 1.9 million genome-wide SNPs, we contrast broad-scale geographical analyses of 10 urban and 11 non-urban locales with fine-scale within-city analyses including 30 urban locales across the western USA. These hierarchical datasets enable us to test hypotheses of how urbanization impacts multiple urban cities and their genetic connectivity at different spatial scales. Coupled fine-scale and broad-scale analyses reveal contrasting patterns of high and low genetic differentiation among locales within cities as a result of low and high genetic connectivity, respectively, of these cities to the overall population network. We discuss these results as they challenge the use of cities as replicates of urban eco-evolution, and have implications for conservation and human health in a rapidly growing urban habitat.

1. Introduction

While the global human population continues to grow, the regions most impacted by this growth are urban areas, where half of the human population already resides [1]. This urban growth fragments and eliminates the surrounding natural habitat, and can have negative effects on local flora and fauna [2,3]. From an eco-evolutionary perspective, the urban fragmentation model of gene flow predicts populations become isolated because natural corridors are fragmented by urbanization, which reduces dispersal and gene flow [4–10]. These isolated patches are vulnerable to increased genetic drift, reduced genetic diversity within patches and increased genetic differentiation among them [3]. However, urban adapters, which thrive in urban habitats and can be pests of human health concern [11,12], may use these novel urban habitats and human transportation as corridors to increase dispersal [13]. In this respect, the alternative urban facilitation model of gene flow predicts populations become more connected because of these artificial corridors [13]. These urban adapters are models for understanding the evolutionary success of urban adaptation juxtaposed with the many conflicting accounts of urbanization; however, there is a paucity of population genetic studies that have focused on these models. Additionally, urban population genetic studies have traditionally focused either on fine-scale, within single-city patterns of gene flow [14–16] or on broad-scale global, between-city patterns [17–19]. A perspective that bridges fine- and broad-scale patterns of urban connectivity is necessary given the growing realization that cities are not replicates of the processes that lead to urban adaptation [20,21].

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Across broad scales, cities have some similarities, such as buildings and impervious surfaces, because they have been designed specifically to meet the needs of humans [2]; however, the extent to which heterogeneity both within and between cities impacts genetic connectivity is unclear [9,10,20]. Urban population genetic studies have been dominated by analyses within temperate ecosystems and in older, developed cities [21], where urban expansion is typically upon the backbone of landscapes that are human-modified [22,23]. However, more than any other region of the USA, urban expansion has been significantly rapid in the west [24]. This urban growth is unique because much of the increased urban area is built upon natural landscapes, and these have the highest increase in human population size in the USA. In fact, the southwestern USA is further unique as an urban model in that growth is in arid regions where supplemental water use and increased artificial urban heat island temperatures greatly impact local biodiversity [25–27]. These new urban models are suitable habitats for invasive urban adapters, and thus, as urban areas continue to rapidly grow, we need to determine how they impact genetic connectivity among urban adapter populations.

In characterizing evolutionary changes on multiple spatial scales, interdisciplinary approaches in landscape, population and evolutionary genetics have emerged that provide measures of how gene flow moves across the network as a whole [28–30]. Additional social network approaches that compare fine- and broad-scale connectivity increase our understanding of how cities act as a biological network with connections that not only fragment but also facilitate gene flow among them [31]. From a classical population genetic perspective, when gene flow is sufficiently high between populations, genetic diversity within individual populations can be maintained at higher levels [32,33]. While previous studies have characterized dispersal patterns of pest species and their associated patterns of genetic diversity [17–19,34], the use of population genetic and social network analyses that specifically identify urban hubs of connectivity, which maintain genetic diversity and stable population structure, are critical for management of both endangered and pest species [35,36].

The western black widow spider, *Latrodectus hesperus*, is an ideal eco-evolutionary model for examining urban gene flow across both fine and broad scales. Our previous population genetic work [37] focused on this organism as it maintains a large geographical distribution across the arid western USA, inhabiting multiple urban and non-urban areas, and most importantly, is an urban adapter and pest with medical-relevance. Specifically, we and our colleagues have previously documented ecological differences between urban and non-urban *L. hesperus* for changes in fertility, behaviour, web-building and diet [38–40], with dense aggregations in urban areas [40], all of which have health concerns given its highly toxic, vertebrate-specific venom [41]. Our previous sampling of thousands of genome-wide mitochondrial and nuclear SNPs from 11 urban and 10 non-urban locales (i.e. sampled geographical locations, which are not necessarily biological 'populations') found urban-specific patterns of higher within-locale genetic diversity, lower between-locale genetic differentiation, and higher genetic connectivity, all of which are predicted by the urban facilitation model of gene flow. Additionally, we found that not all cities are highly connected, with specific urban hubs driving gene flow among historically isolated non-urban locales. While this previous study provided needed support for our understanding of urban facilitation models and urban pest adaptation, as previously noted, it is unclear how this higher gene flow on the urban landscape impacts genetic diversity and gene flow within different cities in the urban network.

Here, we combine fine-scale and broad-scale population genetic and social network analyses to test the hypothesis that levels of genetic diversity within and between locales are similar among urban areas. Alternatively, because we have previously documented patterns of higher population structure associated with non-urban compared to urban locales, and that some urban locales contribute significantly more to overall broad-scale genetic connectivity than others, we predict that cities more connected to the urban network will tend to have higher levels of within-city genetic diversity. We compare and contrast our previous broad-scale patterns of urban gene flow [37] with a new fine-scale locale sampling from within three southwestern US cities. Our application of social network analyses enables us to determine if patterns of fine-scale within-city genetic connectivity are consistent across multiple cities, as well as to determine whether these patterns are predicted by a city's connectivity to the overall network. Results from these population networks have implications for applied urban development, management of endangered and pest species diversity within and across cities, and human health management across different urban areas.

2. Methods

(a) Sampling

The western black widow spider (*L. hesperus*) is a nocturnal web-building predator that is both asocial and highly cannibalistic in all life stages [39]. In urban areas, populations of western black widow spiders are typically densely aggregated in open xeric-landscapes [39], whereas their non-urban distribution is very patchy and isolated associated with arid, rocky-outcrops and dry river-bed banks that are highly sheltered [42]. In considering these distributions and the difficulty in access to western black widow spider habitat, we use our previously published samples [37] across the entire geographical range as our 'broad-scale sample', with an extensive within-city sampling from multiple urban areas as our 'fine-scale sample' added here (figure 1; electronic supplementary material, table S1). Specifically, the broad-scale analyses include the Miles *et al*. [37] data from 11 urban and 10 non-urban locales, with 10 individuals from each locale, for a total of 210 individuals. The fine-scale analyses are composed of three urban areas from the southwestern USA in Albuquerque (NM), Las Vegas (NV) and Phoenix (AZ), and each urban area includes 10 individuals from each of 10 urban locales and 1 non-urban locale, for a total of 330 individuals from the three urban areas. We based this current study on these three urban areas because they each included one urban and one non-urban locale in close proximity from our Miles *et al*. [37] dataset. Thus, with this overlap of 3 urban and 3 non-urban locales between the broad- and fine-scale samples, overall, we have a combined dataset of 480 individuals from 48 (38 urban and 10 non-urban) locales. Each locale included a sample area of approximately 0.5 km^2. Males are significantly smaller and rarely found; thus, our sample is almost all females (fewer than 10 males). All individuals were placed in 90% EtOH and stored at −20°C before DNA extraction.

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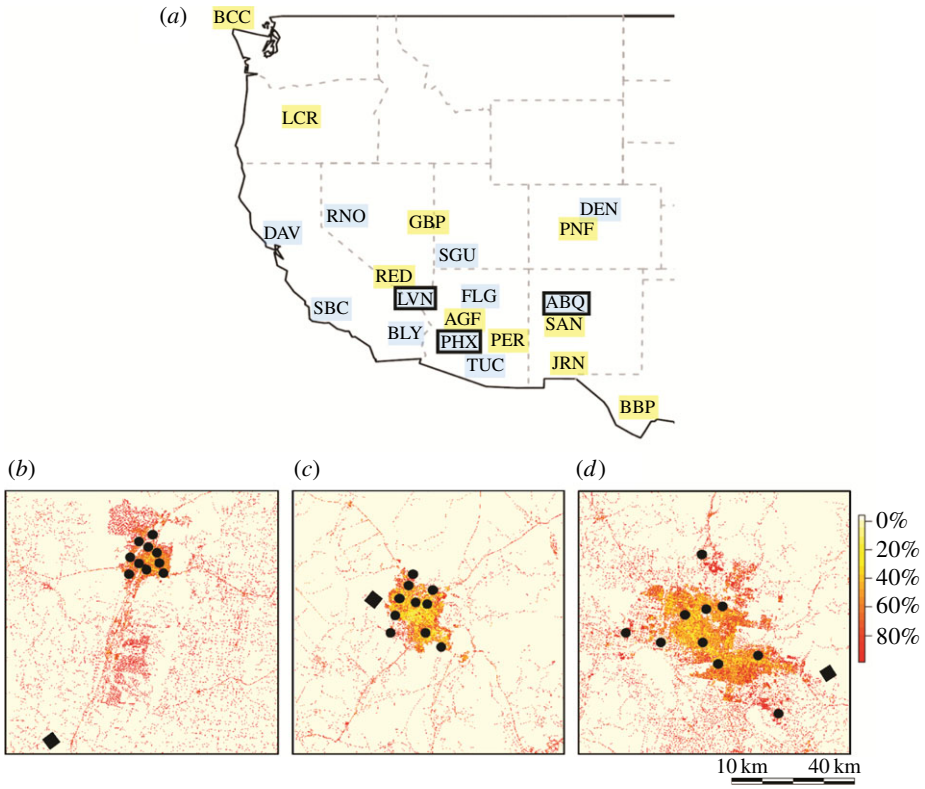

Figure 1. (*a*) Geographical distribution of the broad-scale sampled locales of the western black widow spider across the western USA (see electronic supplementary material, table S2). Highlighted locales in blue and yellow reflect urban and non-urban samples, respectively. Fine-scale sampled locales are shown for (*b*) Albuquerque, (*c*) Las Vegas and (*d*) Phoenix (see electronic supplementary material, table S1 for sampling locales), with circles for urban and diamonds for non-urban locales. Colour scale represents the percent impervious surface. (Online version in colour.)

These three southwestern US cities are similarly located within an arid landscape, with recent human population and geographical size expansion. However, they also have varying urbanization histories for the southwest with respect to colonization time, geographical size and human population size [24], with which to contrast the impact of urbanization on genetic connectivity within urban areas (figure 1). Albuquerque is the smallest and oldest of the three cities founded in 1706, and covers 490 km^2 with a current human population of 560 000; Las Vegas is the most recently founded in 1905, covers 1600 km^2, and is one of the fastest growing metropolitan areas with a population of 1.9 million; Phoenix is the largest of the three, having been founded in 1881, covers 235 000 km^2 and has a population of 4.5 million as the 12th largest metropolitan area in the USA. Although the population size of Albuquerque has remained relatively small, likely due to it being bounded by the Sandia Mountains on the east and Native American land on the west, Phoenix and Las Vegas have been two of the fastest-growing metropolitan areas, expanding over 45% in the last 30 years [24].

(b) Data collection

Genomic DNA was extracted from tissue dissected from one front and one hind leg of each spider using the DNeasy Blood and Tissue Kit (Qiagen, Valencia, CA, USA). We collected genome-wide nuclear DNA (nuDNA) sequence fragments by generating reduced representation, double-digest RAD-sequencing (ddRADseq) libraries according to a previously published protocol [43]. Extracted DNA was digested with MseI and EcoRI (New England Biolabs, Ipswich, MA, USA), ligated with adapters containing Illumina amplification and sequencing primers and unique barcodes [43], and then PCR amplified. Barcoded individuals were pooled (20 per library), then size selected using gel electrophoresis for fragments ranging from 300–500 bp. Fragments were excised and purified using QIAquick Gel Extraction Kits (Qiagen). Each library was sequenced in one lane of an Illumina HiSeq 4000 (150 bp, single end) at the VCU Nucleic Acids Research Facility.

The STACKS v.1.44 *de novo* pipeline [44,45] was used to demultiplex, quality filter, and call genotypes with the following programs' parameters set to default unless otherwise noted: *process_radtags*, *ustacks*, *cstacks*, *sstacks* and *populations*. *process_radtags* demultiplexed reads and filtered for both quality and the presence of barcodes, then trimmed reads to 90 bp in length. The *ustacks* minimum number of reads was set at $m = 5$. The *cstacks* default for the number of mismatches allowed between 'tags' or fragments is 2 or less. We set this value to 5 or less as the high amount of genetic diversity previously reported for *L. hesperus* [37] predicts that multiple fragments would be generated with higher mismatches to the catalog if the default was used. The *populations* minimum coverage was set to 5× per allele for each individual; we retained loci that were sequenced in at least half of the individuals in each locale ($r = 0.5$) and represented in at least 5 locales (approx. 10%; $p = 5$). For analyses not dependent upon estimates of nucleotide site diversity (e.g. population structure), only one SNP per fragment was randomly sampled as a standard way to reduce impact of linkage disequilibrium and selection. Genotype data were exported from STACKS in each of the formats needed for analyses.

(c) Data analysis

Estimates of genetic diversity within and between cities, within and between non-urban locales, and between urban and non-urban locales within urban areas were performed. Estimates of genetic diversity were calculated as the average number of pairwise differences (π), the number of polymorphic sites (S), and the distributions reflecting contrasts between these two (Tajima's

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D [46]). We used standard pairwise F_{ST} measures of overall genetic differentiation between all locales, as well as for measures of genetic differentiation among urban locales within each city to contrast across cities. As in Dyer *et al*. [47], we used a permutation analysis with an in-house script in *R* to test for significant differences in F_{ST}, as these pairwise distributions lack normality and independence. We randomly sampled pairwise F_{ST} values ($n = 1000$) to simulate null distributions of that expected for within-locale and between-locale differentiation. Observed F_{ST} values were compared to these distributions to determine, for example, whether the amount of genetic differentiation among sampled locales within a city was statistically similar across the three cities given the level of genetic differentiation among the three cities.

To examine hierarchical partitioning of genetic variance within and between cities, we performed an AMOVA in R v. 3.4.4 using the *adonis* function in the *vegan* v. 2.5-1 package [48]. To identify potential population clustering, principal component analyses (PCA) were performed in the *gstudio* v. 1.5.0 R package [49]. The PCAs were performed for each of the three urban areas individually including their non-urban locale, as well as for the combined broad-scale and fine-scale samples of a total of 48 locales. We also used the Bayesian clustering algorithm, fastSTRUCTURE [50] with 20 iterations for each *K* cluster. CLUMPAK [51] was used to summarize the output, which was visualized using DISTRUCT [52].

Genetic connectivity among sampled locales was determined from the conditional genetic distance statistic cGD [29], which is estimated from the genetic covariance among locales. This measure of genetic covariance derived from all locales can be visualized as a population graph or 'popgraph' using the *popgraph* v. 1.5 R package [53], where nodes represent sampled locales, and edges represent genetic connections among locales. Popgraph topology is not only a visualization of cGD as genetic covariance but also of social network parameters that define the popgraph. Our application of a social network model evaluates genetic relationships among locales and relative contributions of key 'actors' using mathematical graph theory [31]. This model visually represents gene flow among all sampled locales to identify hubs of higher connectivity on the landscape. Social network node-specific parameters including closeness, degree, betweenness and eigenvector centrality were estimated from popgraphs using the *popgraph* R package. 'Closeness' evaluates the extent to which a locale is genetically similar to all other locales in the network, where higher closeness values indicate higher genetic distance between a locale and other locales in the network. 'Degree' is the absolute number of genetic connections a locale has with other locales in the network, where lower degree values may indicate a locale is more relatively isolated from locales in the network. 'Betweenness' is the sum of the shortest connections, where higher betweenness values indicate a locale has relatively more paths that flow through it from other locales. Finally, eigenvector 'centrality' reflects the extent to which each locale is centrally located as a hub of genetic connectivity, and reflects a sum of the other popgraph parameters here. Altogether, we interpret lower values of 'closeness' and higher values of 'degree', 'betweenness' and 'centrality' as indicative of greater gene flow. To test competing hypotheses, a popgraph was generated for each of the three urban areas of the fine-scale sample, independently, as well as for the combined overall 48 locale dataset. As above, we used permutation analyses to test for significant differences within and among popgraph topologies and parameters within them.

From a spatial perspective, patterns of gene flow, even on a fine spatial scale within cities, may be due to geographical distance when dispersal distance is low, and this is typically expected in web-building spiders [54]. Therefore, we tested a standard isolation-by-distance (IBD) model for each of the three cities. As this analysis is specifically contrasting patterns of gene flow within and across cities, we excluded the non-urban sample from each of the three analyses. Euclidean geographical distances were estimated from latitude and longitude coordinates using the *fields* v. 9.6 R package [55], and genetic distances were calculated as cGD (see above). Mantel tests were performed on the geographical and genetic distances in R. To compare with this analysis of geographical distance, we also used percent impervious surface (PIS) as a standard resistance distance proxy [9,21,23] for the degree of urbanization (national land cover database: https://www.mrlc.gov/finddata.php). As in Dyer *et al*. [47], we performed a permutation analysis using the *gstudio* R package to test for significant relationships between PIS and genetic connectivity. In the permutation analysis, each sampled locale, or node, was fixed on the landscape of the popgraph. The connections among the nodes, or edges of the popgraph, were overlaid on the raster maps of PIS to generate the observed mean and variance resistance distances. We simulated 1000 popgraphs with the observed number of nodes fixed on the landscape and the edges randomized among these popgraphs to generate a null distribution for both the mean and variance of PIS. Statistical significance was assessed by determining the probability of the observed popgraph values compared to our simulated distributions.

3. Results

(a) Within and between city genetic diversity and differentiation

After initial quality filtering, the dataset included 1.9 million SNPs for the broad-scale and fine-scale samples of the overall 48 locales. When additional population parameters in STACKS were applied, greater than 85 000 SNPs were retained with an average read depth of 10× coverage at each locus. Estimates of western black widow spider genetic diversity in the entire sample (electronic supplementary material, table S3) are consistent with estimates of arthropod nuclear genetic diversity in general [56].

For the fine-scale analysis, within-locale estimates of genetic diversity are on average significantly lower for Albuquerque locales ($\pi_{ave} = 0.07\%$) than Las Vegas ($\pi_{ave} = 0.19\%$; $t_{18} = 5.16$, $p < 0.0001$) and Phoenix ($\pi_{ave} = 0.21\%$; $t_{18} = 5.13$, $p < 0.0001$) locales, with Las Vegas and Phoenix having similar estimates ($t_{18} = -1.05$, *ns*). Although estimates of within-city locale genetic differentiation (i.e. among locales within a city) are moderately high, Albuquerque has a significantly higher average pairwise F_{ST} ($F_{ST} = 0.40$, $p < 0.01$; electronic supplementary material, figure S1a), whereas Las Vegas ($F_{ST} = 0.23$, $p < 0.01$; electronic supplementary material, figure S1b) and Phoenix ($F_{ST} = 0.28$, $p < 0.01$; electronic supplementary material, figure S1c) have significantly lower average pairwise F_{ST}. In addition, the average pairwise F_{ST} between Las Vegas and Phoenix ($F_{ST} = 0.26$) is significantly lower than that observed between Albuquerque and each of these two cities (versus Las Vegas, $F_{ST} = 0.41$; versus Phoenix, $F_{ST} = 0.42$, $p < 0.01$). In the combined dataset of 48 locales, urban locales are significantly less genetically differentiated from each other than non-urban locales ($F_{ST} = 0.15$ versus 0.30, $p < 0.01$; electronic supplementary material, figure S2). Each of the three cities has a significantly negative Tajima's *D* value, although these values are not significantly different from each other (electronic supplementary material, table S3). Finally, the

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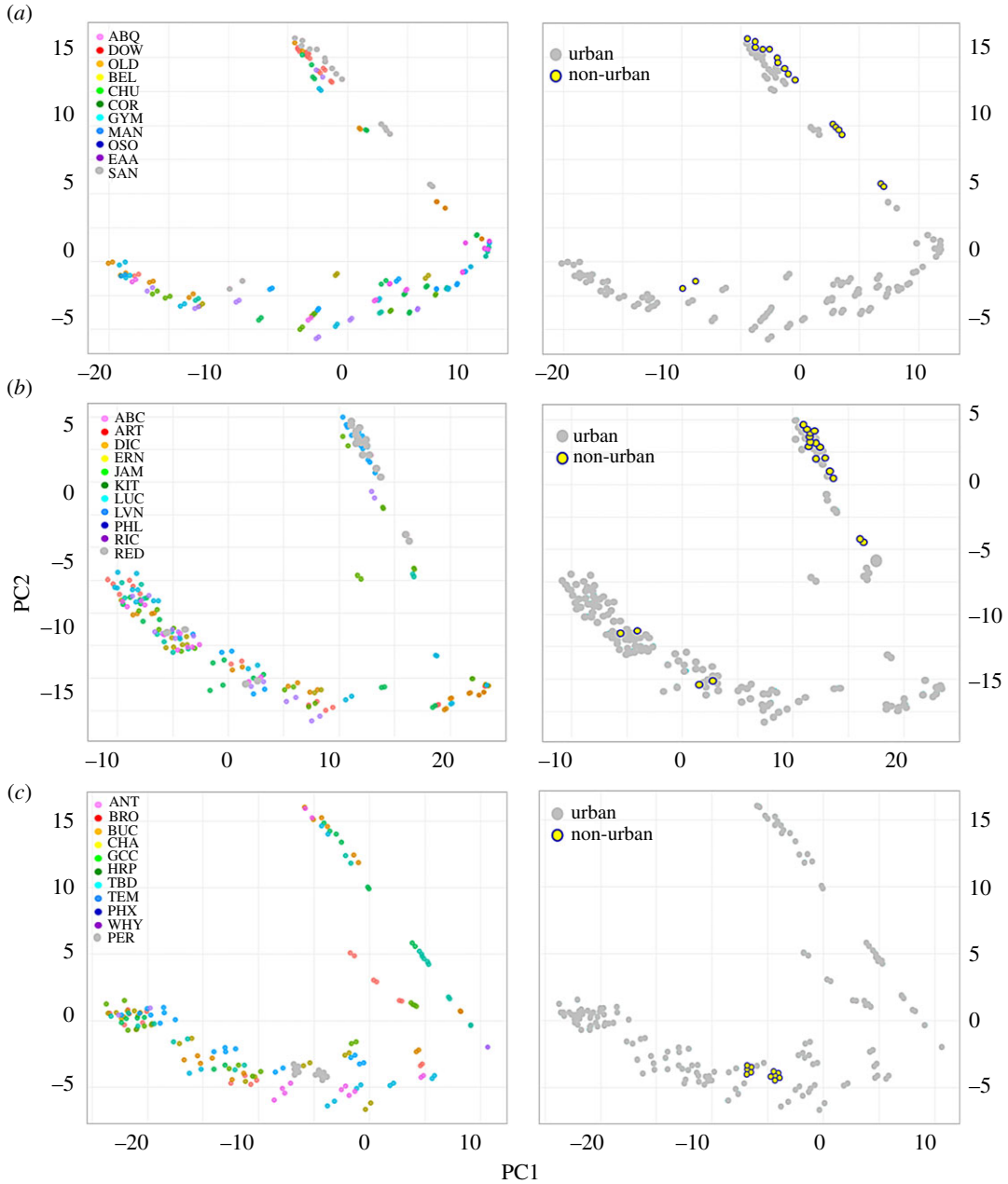

Figure 2. PC1 and PC2 biplots of individual genotypes are shown for fine-scale sampled locales within (*a*) Albuquerque, (*b*) Las Vegas and (*c*) Phoenix urban areas. The left and right panels reflect urban samples highlighted (colour-scheme) and non-urban samples highlighted (yellow), respectively (see electronic supplementary material, table S1). (Online version in colour.)

AMOVA resulted in 8.2% variance explained by city and 20.4% variance explained by locale, with the remaining 71.4% among individuals. That is, the majority of the genetic variance is found among individuals, and overall, more genetic variance is found among locales within cities than is found between cities.

(b) Population structure of urban and non-urban locales

The first 10 PCs for each urban area account for 52% (Albuquerque), 44% (Las Vegas) and 47% (Phoenix) of the genetic variance among individuals (figure 2). The previous PCA of the broad-scale sample had shown significant independent non-urban clusters, with the majority of urban individuals forming a single cluster [37]. With the independent PCAs of the fine-scale samples, we see that each of the three urban areas show a pattern of no clustering of specific urban locales, and apparent clustering for the non-urban individuals, with Phoenix showing the strongest cluster. In the combined PCA of 48 locales, the locales for each of the three cities show some weak clustering in PC1-2 (29% variance explained). Most of the variance among the three cities in PC1-2, as well as that seen for other PCs, is largely explained by Albuquerque locales (electronic supplementary material, figure S3). We could not reject $K = 1$ for any of the fastSTRUCTURE analyses, whether applied to the broad-scale, fine-scale, or individual city samples. Thus, both the PC and fastSTRUCTURE analyses were consistent with each other in finding little evidence of overall population structure or specific population clustering.

(c) Genetic connectivity across urban and non-urban locales

For our popgraph analyses, the number of edges or connections do not significantly differ among the three cities'

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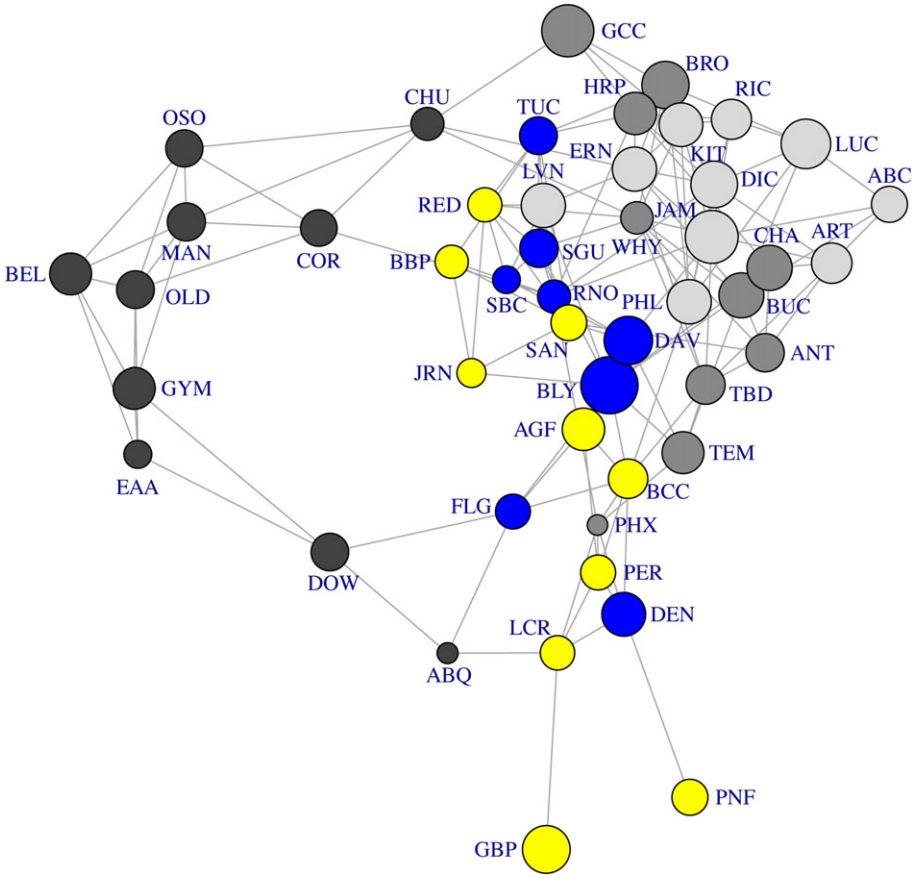

Figure 3. Social network popgraph analysis among urban (blue) and non-urban (yellow) locales for the broad-scale sample (figure 1), as well as the fine-scale sample from Albuquerque (black), Las Vegas (light grey) and Phoenix (dark grey) cities (see electronic supplementary material, tables S1 and S2). The size of each node reflects the locale-specific genetic variance, and the length of the edges is proportional to the conditional genetic distance (cGD; see Methods) between locales. (Online version in colour.)

networks (electronic supplementary material, figure S4). However, the measures of cGD and 'closeness' are statistically significantly higher in Albuquerque and Phoenix than in Las Vegas (electronic supplementary material, table S4). Each of the three cities' popgraphs have contrasting patterns of 'betweenness' such that this parameter is significantly different between all three cities (electronic supplementary material, table S4). Specifically, the Albuquerque popgraph has one locale (BEL) with the highest betweenness value, whereas Las Vegas has nearly each node equally weighted, and Phoenix has two equally weighted locales (BRO and GCC) that have the highest value (electronic supplementary material, table S5). When combining all 48 locales into one popgraph (figure 3), Albuquerque locales have the highest connection distances (least 'central') from all other locales in the network, whereas, Phoenix and Las Vegas locales are centrally connected with all other urban locales in the broad-scale sample. Except for the Albuquerque non-urban locale, non-urban locales are peripherally linked outside of the network and are primarily connected only via urban locales, which make up the core of the overall network. The analysis of 'centrality' identified ERN (Las Vegas), CHU (Albuquerque) and BUC (Phoenix) as the top three major hubs of connectivity in the entire network. On the other hand, non-urban locales have the least influence on connectivity in the overall network; in fact, 7 of the 10 non-urban locales sampled have the lowest 'centrality' of all 48 locales in the network (electronic supplementary material, table S6).

The Mantel tests found both Albuquerque and Phoenix have statistically significant patterns of IBD (Mantel $r^2 = 0.18$ and $r^2 = 0.17$, respectively, both $p < 0.01$), whereas, Las Vegas shows no such pattern (Mantel $r^2 = 0.01$, $p = 0.49$; electronic supplementary material, figure S5). For our PIS resistance distance analyses, the mean and variance for PIS in each of the three cities could not significantly explain the observed popgraph topologies (electronic supplementary material, figure S6), which reflect genetic connectivity among locales within each city.

4. Discussion

Our previous work on the western black widow spider as an urban pest model documented population genetic signatures consistent with the urban facilitation model of gene flow on a broad geographical scale, yet it raised questions about whether this model explains patterns of genetic diversity on fine-scales within cities [37]. Many studies have focused on fine-scale sampling of a single city and its surrounding areas to document genetic diversity and gene flow patterns in testing hypotheses about impacts of urbanization [14–19]. Here, we used a unique analysis of fine-scale sampling of western black widow spider genetic variation from three southwestern US cities in combination with our previous broad-scale urban and non-urban sampling. The primary observation is that urban areas have significantly different patterns of connectivity to the overall network that generate contrasting patterns of within- and between-city genetic diversity for different cities. We discuss these results as they challenge the use of cities as replicates of urban

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eco-evolution, and have implications for conservation and human health in a rapidly growing urban habitat.

(a) Genetic diversity and population structure

The observations of significantly higher within-locale genetic diversity, lower between-locale genetic diversity, and most interestingly, higher connectivity among 11 urban locales compared to 10 non-urban locales were all patterns consistent with the urban facilitation model [10,13] for our previous analysis of western black widow spiders [37]. These patterns are overall consistent with the fine-scaled analyses of 30 locales from three southwestern US cities, whether independently analysed or in combination with the previous broad-scale sample. This result indicated at the outset that broad-scale and fine-scale analyses were not reflecting different general urban evolutionary forces. The various analyses of population structure were consistent with each other, and with the urban facilitation model, in showing no evidence of population clustering at any level. In fact, the hierarchical variance analysis of these broad-scale and fine-scale samples shows that overall, urban-facilitated gene flow both within and among cities results in more genetic differentiation among locales within cities than is found between cities. As we have previously noted [37], these patterns should be expected in emerging studies as signatures of urban-facilitated gene flow for urban adapter and pest species of human health concern [17–19].

(b) Genetic connectivity across the population network

Underlying this urban facilitation model, the most interesting find here is the significant heterogeneity among the fine-scale city samples. Specifically, the locales sampled from within each of Las Vegas and Phoenix show similar levels of within- and between-locale genetic diversity, similar population clustering, and significantly higher genetic connectivity to the overall network. However, Albuquerque has significantly lower within-locale and higher between-locale diversity compared to these same measures in the other two cities. In fact, Albuquerque locales share patterns more in common with the 11 geographically distributed non-urban locales, which appear to have been relatively isolated with lower within-locale diversity and higher population structure [37], and show significantly reduced connectivity to the urban network at large from the popgraph of all 48 locales. Thus, while urban and non-urban areas are different with respect to genetic diversity, even urban areas cannot be classified as a single group with respect to effects of urbanization.

Our previous broad-scale analysis first revealed that certain urban areas act as 'drivers' of the overall higher genetic connectivity of the western black widow spider population network, with surprisingly, even non-urban locales being more connected via urban areas [37]. With the popgraph analysis of the overall 48 locales here, our fine-scale samples are consistent with this initial observation, yet now reveal how urban areas specifically drive connectivity. For example, our social network analysis finds that while Las Vegas and Phoenix locales overall are highly connected to the network, there are multiple locales within Phoenix identified as 'hubs' of connectivity, whereas, locales within Las Vegas each similarly impact gene flow. Alternatively, Albuquerque locales, which overall are significantly more disconnected from the network, includes one hub, and this hub simply connects the other nine Albuquerque locales to the network. Therefore, while certain urban hubs are impacting the network of urban and non-urban locales on the whole, other urban hubs connect only peripheral locales, albeit loosely, to the network. These results reveal one of the powerful characteristics of using conditional genetic distances (cGD) in that the addition or removal of populations alters the covariance across the network [30,57,58], as seen from contrasts of individual-city popgraphs to the overall popgraph. Thus, social network analyses are ideally suited for investigating evolutionary changes across multiple urban environments, in modelling how the applied management of specific urban hubs may alter and possibly create corridors across spatial scales.

Given the underlying urban facilitation model here, we may predict that broad- and fine-scale patterns of urban genetic connectivity predict patterns of within-urban area genetic diversity, which can be a long-term measure of sustainability [2–8]. In testing this hypothesis in the 38 urban locales, we initially find a negative correlation between connectivity (using the parameter betweenness) and genetic diversity. However, this analysis revealed multiple statistical outliers with high genetic diversity that all coincidentally have the lowest measures of connectivity of all 38 locales. In fact, when these three outliers were removed, the correlation became significantly positive ($r^2 = 0.20$, $p < 0.01$). The outlier locales are all from the broad-scale sample (BLY, DAV, DEN; electronic supplementary material, table S3), and reflect different human population and geographical sizes. Thus, this observation reveals that while a proportion of western black widow spider genetic diversity within urban locales can be predicted by connectivity to the network, underlying this correlation is significant heterogeneity among urban areas that reveals multiple 'urban signatures'. More to the point, several designated 'urban' areas (e.g. Albuquerque) mimic even non-urban areas in that they have similarly low levels of genetic diversity and connectivity due to their isolation on the landscape. Thus, while urbanization appears to facilitate gene flow among even geographically distant locales (as evidenced by the IBD results), some urban locales do show the effects of reduced connectivity, further rejecting urban areas as simple replicates of the same urbanization process.

(c) The urban facilitation model of gene flow and its implications

Under an urban facilitation model of gene flow, it has often been proposed that the similarities among cities, such as human population size, canopy cover and human transportation networks, can be dispersal corridors [10,13,21–23]. Our previous investigation of human population size [37] and this study's investigation of PIS as potential drivers of genetic connectivity for broad- and fine-scale samples, respectively, revealed no significant results for the western black widow spider. However, the contrast in patterns of connectivity across scales shown here further emphasizes the importance of identifying corridors and barriers that evolve differently, especially for cities that vary in size, timing and magnitude of human habitation. For example, while PIS was not a significant predictor of within-city connectivity, we note that not only are the PIS distributions different

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among cities, but they do not show a predictable pattern (i.e. cities with high PIS do not have the lowest genetic connectivity). Thus, in continuing to characterize patterns of genetic connectivity among multiple urban areas for multiple organisms, only then will we successfully model how landscape features, which are typically implicated in driving urban gene flow [10,13,21–23], interact both within and across cities.

As a final point of consideration, urban-facilitated gene flow is predicted to sweep an 'urban ecotype' across not only urban areas, but into non-urban areas as well [59,60]. While our previous broad-scale analyses hinted at this speculation, our popgraph network analyses here find that only specific urban locales may have the opportunity to drive and spread phenotypes into specific urban and non-urban locales (i.e. a standard source–sink dynamic). This model would predict that we would see divergent phenotypes between urban and non-urban environments, as well as potentially multiple independent urban and non-urban phenotypes, depending on the strengths of local adaptation and connectivity in the network [61]. For example, our group has already documented western black widow spider behavioural differences between urban and non-urban habitats, where urban spiders are significantly more densely aggregated and are more aggressive towards prey and conspecifics [39,40], as well as gene expression differences among urban habitats related to metabolism and fertility (L.S.M. & B.C.V. 2018, unpublished data). With respect to certain urban pests like the western black widow spider, it is clear that knowledge of the dynamics of the urban network (e.g. hubs) could pave the way for urban management decisions. For example, while variation in certain phenotypes, such as behaviour and egg densities, across cities may imply one set of managing practices, variation across cities in other phenotypes with higher health impact, such as venom type and delivery, would impose far more attention to multiple management plans. Thus, as the field of urban eco-evolution is focused on characterizing the adaptive traits that define invasion into human habitats, it must consider not only how these traits differ from ancestral habitats, but also how multiple urban ecotypes emerge in response to the heterogeneity of urbanization selective pressures on different spatial scales.

Data accessibility. All data are available from the Dryad Digital Repository: http://dx.doi.org/10.5061/dryad.sq85v00 [62].

Authors' contributions. L.S.M. and B.C.V. designed the study, collected field data, carried out the molecular laboratory work, performed analyses and wrote the paper. R.J.D. assisted in analysis design and revision of the paper.

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References

1. United Nations Population Division. 2014 Urbanization. See http://www.un.org/en/development/desa/population/theme/urbanization/
2. McKinney ML. 2006 Urbanization as a major cause of biotic homogenization. *Biol. Conserv.* **127**, 247–260. (doi:10.1016/j.biocon.2005.09.005)
3. Keyghobadi N. 2007 The genetic implications of habitat fragmentation for animals. *Can. J. Zool. Rev. Canadienne De Zoologie* **85**, 1049–1064. (doi:10.1139/Z07-095)
4. Trizio I, Crestanello B, Galbusera P, Wauters L, Tosi G, Matthysen E, Hauffe H. 2005 Geographical distance and physical barriers shape the genetic structure of Eurasian red squirrels (*Sciurus vulgaris*) in the Italian alps. *Mol. Ecol.* **14**, 469–481. (doi:10.1111/j.1365-294X.2005.02428.x)
5. Vandergast AG, Bohonak AJ, Weissman DB, Fisher RN. 2007 Understanding the genetic effects of recent habitat fragmentation in the context of evolutionary history: phylogeography and landscape genetics of a southern California endemic Jerusalem cricket (Orthoptera: Stenopelmatidae: Stenopelmatus). *Mol. Ecol.* **16**, 977–992. (doi:10.1111/j.1365-294X.2006.03216.x)
6. Vandergast AG, Lewallen EA, Deas J, Bohonak AJ, Weissman DB, Fisher RN. 2009 Loss of genetic connectivity and diversity in urban microreserves in a southern California endemic Jerusalem cricket (Orthoptera: Stenopelmatidae: *Stenopelmatus* n. sp *'santa monica'*). *J. Insect Conserv.* **13**, 329–345. (doi:10.1007/s10841-008-9176-z)
7. Allendorf FW, Luikart G. 2007 *Conservation and the genetics of populations*. Malden, MA: Blackwell Publishing.
8. Debinski DM. Holt RD. 2000 A survey and overview of habitat fragmentation experiments. *Conserv. Biol.*, **14**, 342–355. (doi:10.1046/j.1523-1739.2000.98081.x)
9. Storfer A, Murphy MA, Spear SF, Holderegger R, Waits LP. 2010 Landscape genetics: where are we now? *Mol. Ecol.* **19**, 3496–3514. (doi:10.1111/j.1365-294X.2010.04691.x.)
10. Holderegger R, Di Giulio M. 2010 The genetic effects of roads: a review of empirical evidence. *Basic Appl. Ecol.* **11**, 522–531. (doi:10.1016/j.baae.2010.06.006)
11. Blair RB. 1996 Land use and avian species diversity along an urban gradient. *Ecol. Appl.* **6**, 506–519.
12. Shochat E. 2004 Credit or debit? Resource input changes population dynamics of city-slicker birds. *Oikos* **106**, 622–626.
13. Crispo E, Moore J, Lee-Yaw JA, Gray SM, Haller BC. 2011 Broken barriers: human-induced changes to gene flow and introgression in animals. *Bioessays* **33**, 508–518. (doi:10.1002/bies.201000154)
14. Munshi-South J. 2012 Urban landscape genetics: canopy cover predicts gene flow between white-footed mouse (*Peromyscus leucopus*) populations in New York City. *Mol. Ecol.* **21**, 1360–1378.
15. Munshi-South J, Zak Y, Pehek E. 2013 Conservation genetics of extremely isolated urban populations of the northern dusky salamander (*Desmognathus fuscus*) in New York City. *Peer J.* **1**, e64. (doi: 10.7717/peerj.64)
16. Munshi-South J, Karchenko K. 2010 Rapid, Pervasive genetic differentiation of urban white-footed mouse (*Peromyscus leucopus*) populations in New York City. *Mol. Ecol.* **19**, 4242–4254.
17. Booth W, Balvín O, Vargo EL, Vilímová J, Schal C. 2015 Host association drives genetic divergence in the bed bug, *Cimex lectularius*. *Mol. Ecol.* **24**, 980–992. (doi:10.1111/mec.13086)
18. Booth W, Saenz VL, Santangelo RG, Wang C, Schal C, Vargo EL. 2012 Molecular markers reveal infestation dynamics of the bed bug (Hemiptera: Cimicidae) within apartment buildings. *J. Med. Entomol.* **49**, 535–546.
19. Vargo EL, Crissman JR, Booth W, Santangelo RG, Mukha DV, Schal C. 2014 Hierarchical genetic analysis of German Cockroach (*Blattella germanica*) populations from within buildings to across continents. *PLoS ONE* **9**, e102321. (doi:10.1371/journal.pone.0102321)
20. Grimm NB, Faeth SH, Golubiewski NE, Redman CL, Wu J, Bai X, Briggs JM. 2008 Global change and the ecology of cities. *Science* **319**, 756–760. (doi:10.1126/science.1150195pmid:18258902)
21. Johnson MTJ, Munshi-South J. 2017 The evolution of life in urban environments. *Science* **358**, 607. (doi:10.1126/science.aam8327)

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22. McDonnell MJ, Hahs AK. 2015 Adaptation and adaptedness of organisms to urban environments. *Annu. Rev. Ecol. Evol. Syst.* **46**, 261–280. (doi:10.1146/annurev-ecolsys-112414-054258)
23. Alberti M. 2015 Eco-evolutionary dynamics in an urbanizing planet. *Trends Ecol. Evol.* **30**, 114–126. (doi:10.1016/j.tree.2014.11.007)
24. US Census Bureau. 2016 US Census. See http://www.census.gov
25. Bateman HL, Stromberg JC, Banville MJ, Makings E, Scott BD, Suchy A, Wolkis D. 2015 Novel water sources restore plant and animal communities along an urban river. *Ecohydrology* **8**, 792–811. (doi: 10.1002/eco.1560.)
26. Wang C, Myint SW, Wang Z, Song J. 2016 Spatio-temporal modeling of the urban heat island in the Phoenix metropolitan area: Land use change implications. *Remote Sens.* **8**, 185. (doi: 10.3390/rs8030185.)
27. Chow WT, Salamanca F, Georgescu M, Mahalov A, Milne JM, Ruddell BL. 2014 A multi-method and multi-scale approach for estimating city-wide anthropogenic heat fluxes. *ASTM Spec. Tech. Publ.* **99**, 64–76. (doi: 10.1016/j.atmosenv.2014.09.053.)
28. Dyer RJ. 2015 Population Graphs and Landscape Genetics. *Ann. Rev. Ecol. Evol. Syst.* **46**, 327–342.
29. Dyer RJ, Nason JD. 2004 Population graphs: the graph theoretic shape of genetic structure. *Mol. Ecol.* **13**, 1713–1727.
30. Dyer RJ, Nason JD, Garrick RC. 2010. Landscape modelling of gene flow: Improved power using conditional genetic distance derived from the topology of population networks. *Mol. Ecol.* **19**, 3746–3759.
31. Wasserman S, Faust K. 1994 *Social network analysis: methods and applications*. Cambridge, UK: Cambridge University Press.
32. Dobzhansky T. 1937 *Genetics and the origin of species*. New York, NY: Columbia University Press.
33. Wright SW. 1982 The shifting balance theory and macroevolution. *Annu. Rev. Genet.* **16**, 1–19.
34. Crissman JR, Booth W, Santangelo RG, Mukha DV, Vargo EL, Schal C. 2010 Population genetic structure of the German cockroach (Blattodea: Blattellidae) in apartment buildings. *J. Med. Entomol.* **47**, 553–564.
35. Paupy C, Brengues C, Kamgang B, Hervé JP, Fontenille D, Simard F. 2008 Gene flow between domestic and sylvan populations of *Aedes aegypti* (Diptera: Culicidae) in North Cameroon. *J. Med. Entomol.* **45**, 391–400.
36. Piccinali RV, Marcet PL, Noireau F, Kitron U, Gürtler RE, Dotson EM. 2009 Molecular population genetics and phylogeography of the Chagas disease vector *Triatoma infestansin* South America. *J. Med. Entomol.* **46**, 796–809.
37. Miles LS, Johnson JC, Dyer RJ, Verrelli BC. In press. Urbanization as a facilitator of gene flow in a human health pest. *Mol. Ecol.* (doi: 10.1111/mec.14783)
38. Johnson JC, Trubl PJ, Miles LS. 2012 Black widows in an urban desert: city-living compromises spider fecundity and egg investment despite urban prey abundance. *Am. Midland Nat.* **168**, 333–340.
39. Johnson JC, Miles LS, Trubl PJ, Hagenmaier A. 2014 Maternal effects on egg investment and offspring performance in black widow spiders. *Anim. Behav.* **91**, 67–73. (doi:10.1016/j.anbehav.2014.02.031)
40. Trubl P, Gburek T, Miles L, Johnson JC. 2012 Black widow spiders in an urban desert: population variation in an arthropod pest across metropolitan phoenix, AZ. *Urban Ecosyst.* **15**, 599–609. (doi:10.1007/s11252-011-0223-2)
41. Vetter RS, Isbister GK. 2008 Medical aspects of spider bites. *Annu. Rev. Entomol.* **53**, 409–429.
42. Chamberlin RV, Ivie W. 1935 The black widow spider and its variations in the United States. *Bull. Univ. Utah* **25**, 1–18.
43. Peterson BK, Weber JN, Kay EH, Fisher HS, Hoekstra HE. 2012 Double digest RADseq: an inexpensive method for de novo SNP discovery and genotyping in model and non-model species. *PLoS ONE* **7**, e37135. (doi:10.1371/journal.pone.0037135)
44. Catchen JM, Amores A, Hohenlohe P, Cresko W, Postlethwait JH. 2011 Stacks: building and genotyping loci de novo from short-read sequences. *G3: Genes, Genomes, Genetics* **1**, 171–182. (doi:10.1534/g3.111.000240)
45. Catchen J, Hohenlohe PA, Bassham S, Amores A, Cresko WA. 2013 Stacks: an analysis tool set for population genomics. *Mol. Ecol.* **22**, 3124–3140. (doi:10.1111/mec.12354)
46. Tajima F. 1989 Statistical-method for testing the neutral mutation hypothesis by DNA polymorphism. *Genetics* **123**, 585–595.
47. Dyer RJ, Chan DM, Gardiakos VA, Meadows CA. 2012 Pollination Graphs: quantifying pollen pool covariance networks and the influence of intervening landscape on genetic connectivity in the North American understory tree, *Cornus florida* L. *Landscape Ecol.* **27**, 239–251. (doi:10.1007/s10980-011-9696-x)
48. Oksanen J *et al*. 2017 vegan: community ecology package. R package version 2.4–3. See https://CRAN.R-project.org/package=vegan
49. Dyer RJ. 2016 gstudio: tools related to the spatial analysis of genetic marker data. R package version 1, 0.
50. Raj A, Stephens M, Pritchard JK. 2014 fastSTRUCTURE: variational inference of population structure in large SNP data sets. *Genetics* **197**, 573–589.
51. Kopelman NM, Mayzel J, Jakobsson M, Rosenberg NA, Mayrose I. 2015 Clumpak: a program for identifying clustering modes and packaging population structure inferences across K. *Mol. Ecol. Resour.* **15**, 1179–1191.
52. Rosenberg NA. 2004 Distruct: a program for the graphical display of population structure. *Mol. Ecol. Notes* **4**, 137–138.
53. Dyer RJ. 2017 popgraph: This is an R package that constructs and manipulates population graphs. R package version 1, 0.
54. Foelix RF. 2014 *Biology of spiders*. New York, NY: Oxford University Press.
55. Nychka D, Furrer R, Paige J, Sain S. 2015 fields: tools for spatial data. (doi:10.5065/D6W957CT)
56. Burns M, Starrett J, Derkarabetian S, Richart CH, Cabrero A, Hedin M. 2017 Comparative performance of double-digest RAD sequencing across divergent arachnid lineages. *Mol. Ecol. Resour.* **17**, 418–430. (doi:10.1111/1755-0998.12575)
57. Naujokaitis-Lewis IR, Curtis JMR, Tischendorf L, Badzinski D, Lindsay K, Fortin MJ. 2013 Uncertainties in coupled species distribution–metapopulation dynamics models for risk assessments under climate change. *Divers. Distrib.* **19**, 541–554. (doi:10.1111/ddi.12063)
58. Koen EL, Bowman J, Garroway CJ, Wilson PJ. 2013 The sensitivity of genetic connectivity measures to unsampled and under-sampled sites. *PLoS ONE* **8**, e56204. (https://doi.org/10.1371/journal.pone.0056204)
59. Krtinic B, Ludoski J, Milankov V. 2012. Study on siphonal measurements and usefulness in delimitation of rural and urban ecotypes of *Culex pipiens*. *Bull. Insectol.* **65**, 23.
60. Schapira A, Boutsika K. 2012 Malaria ecotypes and stratification. *Adv. Parasitol.* **78**, 97.
61. Thompson KA, Renaudin M, Johnson MTJ. 2016 Urbanization drives the evolution of parallel clines in plant populations. *Proc. R. Soc. B* **283**. 20162180. (doi:10.1098/rspb.2016.2180)
62. Miles LS, Dyer RJ, Verrelli BC. 2018 Data from: Urban hubs of connectivity: contrasting patterns of gene flow within and among cities in the western black widow spider. Dryad Digital Repository. (doi:10.5061/dryad.sq85v00)